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Gas sensor

Abstract

A gas sensor detects a specific gas concentration in a measurement-object gas and includes an element body and one or more pump cells. The element body includes an oxygen-ion-conductive solid electrolyte layer and is provided with a measurement-object gas flow section therein. The measurement-object gas flow section receives a measurement-object gas and allows the measurement-object gas to flow therethrough. The one or more pump cells each have an inner electrode and an outer pump electrode and pump out oxygen from around the inner electrode to around the outer pump electrode. The inner electrode is disposed in the measurement-object gas flow section and contains a catalytically-active noble metal. At least one pump cell of the one or more pump cells pumps out the oxygen by applying a repeatedly on-off controlled pump current between a measurement electrode and the outer pump electrode.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation application of PCT/JP2020/013415, filed on Mar. 25, 2020, which claims the benefit of priority of Japanese Patent Application No. 2019-064322, filed on Mar. 28, 2019, the entire contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to gas sensors.

2. Description of the Related Art

(2) A known gas sensor in the related art detects the concentration of a specific gas, such as NO_x, in a measurement-object gas, such as exhaust gas of an automobile. For example, Patent Literature 1 describes a gas sensor including a layered body of a plurality of oxygen-ion-conductive solid electrolyte layers, a main pump cell, and a measurement pump cell. The main pump cell includes an inner pump electrode disposed in a measurement-object gas flow section inside the layered body, and also includes an outer pump electrode disposed outside the layered body. The measurement pump cell includes a measurement electrode disposed in the measurement-object gas flow section, and also includes an outer pump electrode. When this gas sensor is to detect the NO_x concentration, the main pump cell first pumps out oxygen from the measurement-object gas flow section or pumps oxygen into the measurement-object gas flow section, so that the oxygen concentration in the measurement-object gas flow section is adjusted. Subsequently, after the oxygen concentration is adjusted, NO_x in the measurement-object gas is reduced, and oxygen produced as a result of the reduction is pumped out by the measurement pump cell. Based on a pump current flowing through the measurement pump cell, the NO_x concentration in the

measurement-object gas is detected.

CITATION LIST

Patent Literature

(3) PTL 1: JP 2015-200642 A

SUMMARY OF THE INVENTION

(4) Each of the electrodes included in the pump cells of the gas sensor is catalytically active (i.e., has catalytic performance) and serves as a catalyst for a reaction where oxygen turns into ions. However, as the gas sensor is used, the catalytic activity of this electrode may change, as in a case where the catalytic activity decreases (i.e., is deactivated). A change in the catalytic activity may sometimes lead to a problem in the gas sensor, such as deterioration in the sensitivity for detecting the specific gas concentration.

(5) The present invention has been made to solve such a problem, and a main object thereof is to suppress a change in the catalytic activity of an electrode due to use of the gas sensor.

(6) The present invention provides the following solutions for achieving the main object mentioned above.

(7) A gas sensor according to the present invention detects a specific gas concentration in a measurement-object gas and includes an element body and one or more pump cells. The element body includes an oxygen-ion-conductive solid electrolyte layer and is provided with a measurement-object gas flow section therein. The measurement-object gas flow section receives a measurement-object gas and allows the measurement-object gas to flow therethrough. The one or more pump cells each have an inner electrode and an outer electrode and pump out oxygen from around the inner electrode to around the outer electrode. The inner electrode is disposed in the measurement-object gas flow section and contains a catalytically-active noble metal. The outer electrode is disposed in an area to be exposed to the measurement-object gas at an outer side of the element body. At least one pump cell of the one or more pump cells pumps out the oxygen by applying a repeatedly on-off controlled pump current between the inner electrode and the outer electrode.

(8) This gas sensor includes the one or more pump cells. At least one of the pump cells pumps out the oxygen from around the inner electrode to around the outer electrode by applying the repeatedly on-off controlled pump current between the inner electrode and the outer electrode. By applying the repeatedly on-off controlled pump current (referred to as “intermittent pump current” hereinafter) in this manner, a change in the catalytic activity of the inner electrode due to use of the gas sensor can be suppressed, as compared with a case where a continuous pump current is applied to a pump cell. A conceivable reason for this will be described below. When oxygen is to be pumped out from around the inner electrode by applying the pump current between the inner electrode and the outer electrode, the surrounding oxygen turns into oxygen ions inside the inner electrode. The oxygen ions serve as electron carriers and travel toward the outer electrode. During this process, if the pump current is a continuous electric current, a reaction where some noble metals in the inner electrode oxidize and a reaction where oxygen ions are released as a result of reduction of the oxidized noble metals both occur. When both reactions reach a state of equilibrium, some of the noble metals in the inner electrode are in a constantly oxidized state. Since oxidized noble metals tend to evaporate more easily than before they are oxidized, the noble metals in the inner electrode tend to decrease with use of the gas sensor, thus causing the catalytic activity of the inner electrode to change. On the other hand, the following description relates to a case where an intermittent pump current is used and is intermittently applied such that the intermittent pump current becomes an average electric current equal to the continuous pump current. In this case, the pump current applied to the inner electrode during an on mode is a value larger than the continuous pump current. Accordingly, the oxygen inside the inner electrode turns into ions and travel toward the outer electrode more during the on mode of the intermittent pump current than when the pump current is applied continuously, so that the oxygen concentration inside the inner

electrode decreases. In such a state where the oxygen concentration inside the inner electrode is low, the aforementioned noble metals in the inner electrode are less likely to oxidize, and rather, reduction of oxidized noble metals may occur. Thus, a decrease in the aforementioned noble metals in the inner electrode with use of the gas sensor is suppressed. Furthermore, electric current hardly flows to the inner electrode when the pump current is in an off mode, so that oxidization of the aforementioned noble metals in the inner electrode is less likely to occur. As a result, oxidization of the noble metals in the inner electrode is suppressed in both an on mode and an off mode. It is thus conceivable that a change in the catalytic activity of the inner electrode due to use of the gas sensor is suppressed.

(9) In the gas sensor according to the present invention, the one or more pump cells may include a main pump cell, an auxiliary pump cell, and a measurement pump cell. The main pump cell has an inner main pump electrode serving as the inner electrode and an outer main pump electrode serving as the outer electrode. The inner main pump electrode is disposed in a first internal cavity in the measurement-object gas flow section. The main pump cell pumps out oxygen from the first internal cavity. The auxiliary pump cell has an inner auxiliary pump electrode serving as the inner electrode and an outer auxiliary pump electrode serving as the outer electrode. The inner auxiliary pump electrode is disposed in a second internal cavity provided downstream of the first internal cavity in the measurement-object gas flow section. The auxiliary pump cell pumps out oxygen from the second internal cavity. The measurement pump cell has an inner measurement electrode serving as the inner electrode and an outer measurement electrode serving as the outer electrode. The inner measurement electrode is disposed in a measurement chamber provided downstream of the second internal cavity in the measurement-object gas flow section. The measurement pump cell pumps out oxygen produced in the measurement chamber from the specific gas. At least one of the main pump cell, the auxiliary pump cell, and the measurement pump cell pumps out the oxygen from around the inner electrode by applying the repeatedly on-off controlled pump current between the inner electrode and the outer electrode. Accordingly, with regard to a pump cell that is caused to operate by receiving an intermittent pump current among the main pump cell, the auxiliary pump cell, and the measurement pump cell, a change in the catalytic activity of the inner electrode due to use of the gas sensor can be suppressed.

(10) In this case, the measurement pump cell may pump out the oxygen by applying a measurement pump current serving as the repeatedly on-off controlled pump current between the inner measurement electrode and the outer measurement electrode. In other words, among the main pump cell, the auxiliary pump cell, and the measurement pump cell, at least the measurement pump cell may be caused to operate by using an intermittent pump current. Accordingly, a change in the catalytic activity of the inner measurement electrode due to use of the gas sensor can be suppressed. A change in the catalytic activity of the inner measurement electrode has a greater effect on the sensitivity for detecting the specific gas concentration in the gas sensor than a change in the catalytic activity of the inner main pump electrode and the inner auxiliary pump electrode. Therefore, deterioration in the detection sensitivity may be readily minimized by suppressing a change in the catalytic activity of the inner measurement electrode.

(11) In this case, the gas sensor according to the present invention may further include a reference electrode, a measurement-voltage detection device, a measurement-pump-cell control device, and a specific-gas-concentration detection device. The reference electrode is disposed inside the element body and receives a reference gas serving as a reference for detection of the specific gas concentration. The measurement-voltage detection device detects a measurement voltage between the reference electrode and the inner measurement electrode. The measurement-pump-cell control device controls the measurement pump current based on the measurement voltage during a second period so as to set an oxygen concentration in the measurement chamber to a predetermined low concentration. The second period is one of a first period in which a change has occurred in the measurement voltage due to the measurement pump current being in an on mode and a second

period in which the change in the measurement voltage has receded as compared with the first period due to the measurement pump current being in an off mode. The specific-gas-concentration detection device detects the specific gas concentration in the measurement-object gas based on the measurement pump current. The effect that the measurement pump current has on the measurement voltage during the second period is smaller than during the first period. Therefore, the oxygen concentration in the measurement chamber is adjusted by controlling the measurement pump current based on the measurement voltage during the second period, and the specific gas concentration in the measurement-object gas is detected based on the measurement pump current, whereby the specific gas concentration can be accurately detected.

(12) In the example where the measurement pump cell pumps out the oxygen by applying the repeatedly on-off controlled measurement pump current, the gas sensor according to the present invention may further include a reference electrode, a measurement-voltage detection device, a measurement-pump-cell control device, and a specific-gas-concentration detection device. The reference electrode is disposed inside the element body and receives a reference gas serving as a reference for detection of the specific gas concentration. The measurement-voltage detection device detects a measurement voltage between the reference electrode and the inner measurement electrode. The measurement-pump-cell control device controls the measurement pump current so as to set an average value of the measurement pump current to a predetermined target value. The specific-gas-concentration detection device detects the specific gas concentration in the measurement-object gas based on the measurement voltage during a second period. The second period is one of a first period in which a change has occurred in the measurement voltage due to the measurement pump current being in an on mode and a second period in which the change in the measurement voltage has receded as compared with the first period due to the measurement pump current being in an off mode. Accordingly, the specific gas concentration in the measurement-object gas can be detected. Moreover, the specific gas concentration in the measurement-object gas is detected based on the measurement voltage during the second period, whereby the specific gas concentration can be accurately detected.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic cross-sectional view of a gas sensor **100**.
- (2) FIG. 2 is a block diagram illustrating an electrical connection relationship between a controller **90** and individual cells.
- (3) FIG. 3 illustrates temporal changes in a pump current I_{p2} and a voltage $V2$.
- (4) FIG. 4 is a schematic cross-sectional view of a gas sensor **900** according to a comparative example.
- (5) FIG. 5 illustrates a temporal change in the pump current I_{p2} of the gas sensor **900**.
- (6) FIG. 6 illustrates an example where the pump current I_{p2} is a burst pulse.
- (7) FIG. 7 is a schematic cross-sectional view of a gas sensor **200** according to a modification.
- (8) FIG. 8 is a schematic cross-sectional view of a sensor element **301** according to a modification.
- (9) FIG. 9 illustrates a comparison between pulse currents having different pulse widths.
- (10) FIG. 10 illustrates an example of comparison results between a case where control for reducing the number of pulses is not performed and a case where such control is performed.

DETAILED DESCRIPTION OF THE INVENTION

(11) Embodiments of the present invention will now be described with reference to the drawings. FIG. 1 is a schematic cross-sectional view schematically illustrating an example of the configuration of a gas sensor **100** according to an embodiment of the present invention. FIG. 2 is a block diagram illustrating an electrical connection relationship between a controller **90** and

individual cells. This gas sensor **100** is attached to a pipe, such as an exhaust gas pipe of an internal combustion engine. The gas sensor **100** detects the concentration of a specific gas, such as NO_x or ammonia, in a measurement-object gas, such as exhaust gas of the internal combustion engine. In this embodiment, the gas sensor **100** measures the NO_{sub.x} concentration as the specific gas concentration. The gas sensor **100** includes a sensor element **101** having an elongated rectangular parallelepiped shape, individual cells **21**, **41**, **50**, and **80** to **83** each including a part of the sensor element **101**, and the controller **90** that controls the entire gas sensor **100**.

(12) The sensor element **101** is an element including a layered body in which six layers, namely a first substrate layer **1**, a second substrate layer **2**, a third substrate layer **3**, a first solid electrolyte layer **4**, a spacer layer **5**, and a second solid electrolyte layer **6**, are layered in this order from the bottom side, as viewed in the drawing. Each of the six layers is formed of an oxygen-ion-conductive solid electrolyte layer containing, for example, zirconia (ZrO₂). The solid electrolyte forming these six layers is dense and gastight. This sensor element **101** is manufactured, for example, by stacking ceramic green sheets corresponding to the individual layers on top of each other, for example, after predetermined processing and circuit pattern printing, and then firing the stacked ceramic green sheets so that they are combined together.

(13) A gas inlet **10**, a first diffusion control section **11**, a buffer space **12**, a second diffusion control section **13**, a first internal cavity **20**, a third diffusion control section **30**, a second internal cavity **40**, a fourth diffusion control section **60**, and a third internal cavity **61** are formed adjacent to each other so as to communicate in the above order between the lower surface of the second solid electrolyte layer **6** and the upper surface of the first solid electrolyte layer **4** on the front end side (on the left end side in FIG. **1**) of the sensor element **101**.

(14) The gas inlet **10**, the buffer space **12**, the first internal space **20**, the second internal space **40**, and the third internal space **61** constitute a space within the sensor element **101**. The space is provided in such a manner that a portion of the spacer layer **5** is hollowed out. The top of the space is defined by the lower surface of the second solid electrolyte layer **6**, the bottom of the space is defined by the upper surface of the first solid electrolyte layer **4**, and sides of the space are defined by side surfaces of the spacer layer **5**.

(15) The first diffusion control section **11**, the second diffusion control section **13**, and the third diffusion control section **30** are each provided as two laterally elongated slits (i.e., the longitudinal direction of the openings is perpendicular to the figure). The fourth diffusion control section **60** is provided as a single laterally elongated slit (i.e., the longitudinal direction of the opening is perpendicular to the figure) formed as a clearance under the lower surface of the second solid electrolyte layer **6**. The section extending from the gas inlet **10** to the third internal cavity **61** is also referred to as “measurement-object gas flow section”.

(16) A reference gas introduction space **43** is disposed between the upper surface of the third substrate layer **3** and the lower surface of the spacer layer **5** at a position farther away from the front end side than the measurement-object gas flow section. The reference gas introduction space **43** is defined at both sides by the side surfaces of the first solid electrolyte layer **4**. As an example of a reference gas for NO_x concentration measurement, air is introduced into the reference gas introduction space **43**.

(17) An air introduction layer **48** is a porous ceramic layer. The reference gas is introduced into the air introduction layer **48** through the reference gas introduction space **43**. The air introduction layer **48** is formed so as to cover a reference electrode **42**.

(18) The reference electrode **42** is formed between the upper surface of the third substrate layer **3** and the first solid electrolyte layer **4**. As described above, the air introduction layer **48** leading to the reference gas introduction space **43** is disposed around the reference electrode **42**. As described later, the reference electrode **42** can be used to measure the oxygen concentrations (oxygen partial pressures) in the first internal cavity **20**, the second internal cavity **40**, and the third internal cavity **61**. The reference electrode **42** is formed as a porous cermet electrode (e.g., a cermet electrode

composed of Pt and ZrO.sub.2).

(19) The gas inlet **10** of the measurement-object gas flow section is exposed to the external space. The measurement-object gas is taken into the sensor element **101** through the gas inlet **10** from the external space. The first diffusion control section **11** applies a predetermined diffusion resistance to the measurement-object gas taken in through the gas inlet **10**. The buffer space **12** is provided for guiding the measurement-object gas introduced by the first diffusion control section **11** to the second diffusion control section **13**. The second diffusion control section **13** applies a predetermined diffusion resistance to the measurement-object gas introduced to the first internal cavity **20** from the buffer space **12**. When the measurement-object gas is to be introduced to the first internal cavity **20** from outside the sensor element **101**, the measurement-object gas quickly taken into the sensor element **101** through the gas inlet **10** due to pressure fluctuations of the measurement-object gas in the external space (i.e., pulsations of exhaust pressure if the measurement-object gas is exhaust gas of an automobile) is not directly introduced to the first internal cavity **20** but is introduced to the first internal cavity **20** after the pressure fluctuations of the measurement-object gas are negated through the first diffusion control section **11**, the buffer space **12**, and the second diffusion control section **13**. Accordingly, the pressure fluctuations of the measurement-object gas to be introduced to the first internal cavity **20** can be made substantially negligible. The first internal cavity **20** is provided as a space for adjusting the oxygen partial pressure in the measurement-object gas introduced via the second diffusion control section **13**. The oxygen partial pressure is adjusted by actuating a main pump cell **21**.

(20) The main pump cell **21** is an electrochemical pump cell composed of an inner pump electrode **22** having a ceiling electrode portion **22a** disposed over substantially an entire portion of the lower surface of the second solid electrolyte layer **6** that faces the first internal cavity **20**, an outer pump electrode **23** disposed on a region of the upper surface of the second solid electrolyte layer **6** that corresponds to the ceiling electrode portion **22a** so as to be exposed to the external space, and a portion of the second solid electrolyte layer **6** that is located between the inner pump electrode **22** and the outer pump electrode **23**.

(21) The inner pump electrode **22** is formed on portions of the upper and lower solid electrolyte layers (the second solid electrolyte layer **6** and the first solid electrolyte layer **4**) that define the first internal cavity **20** and portions of the spacer layer **5** that give the sidewalls of the first internal cavity **20**. Specifically, the ceiling electrode portion **22a** is formed on a portion of the lower surface of the second solid electrolyte layer **6** that gives the ceiling surface of the first internal cavity **20**. A bottom electrode portion **22b** is formed on a portion of the upper surface of the first solid electrolyte layer **4** that gives the bottom surface of the first internal cavity **20**. Side electrode portions (not shown) are formed on portions of the sidewall surfaces (inner surfaces) of the spacer layer **5** that form both sidewalls of the first internal cavity **20** so as to join together the ceiling electrode portion **22a** and the bottom electrode portion **22b**. Thus, the inner pump electrode **22** is provided as a tunnel-like structure in the area where the side electrode portions are disposed.

(22) The inner pump electrode **22** contains a catalytically-active noble metal (e.g., at least one of Pt, Rh, Pd, Ru, and Ir). The inner pump electrode **22** also contains a noble metal (e.g., Au) having a catalytic-activity inhibition ability for inhibiting the catalytic activity of the catalytically-active noble metal against the specific gas. Accordingly, the inner pump electrode **22** that comes into contact with the measurement-object gas is composed of a material with a weakened reduction ability against the specific gas (i.e., NOx) component of the measurement-object gas. The inner pump electrode **22** is preferably formed of a cermet containing a noble metal and an oxygen-ion-conductive oxide (i.e., ZrO.sub.2). Moreover, the inner pump electrode **22** is preferably porous. In this embodiment, the inner pump electrode **22** is a porous cermet electrode composed of Pt and ZrO.sub.2 and containing 1% Au.

(23) Similar to the inner pump electrode **22**, the outer pump electrode **23** contains a catalytically-active noble metal. Similar to the inner pump electrode **22**, the outer pump electrode **23** may

contain a noble metal having a catalytic-activity inhibition ability, and may be formed of a cermet. The outer pump electrode **23** is preferably porous. In this embodiment, the outer pump electrode **23** is a porous cermet electrode composed of Pt and ZrO.sub.2.

(24) In the main pump cell **21**, the desired pump voltage V_{p0} is applied between the inner pump electrode **22** and the outer pump electrode **23** so that a pump current I_{p0} flows between the inner pump electrode **22** and the outer pump electrode **23** in either a positive or negative direction. Thus, oxygen can be pumped from the first internal cavity **20** to the external space or from the external space to the first internal cavity **20**.

(25) To detect the oxygen concentration (oxygen partial pressure) in the atmosphere in the first internal cavity **20**, the inner pump electrode **22**, the second solid electrolyte layer **6**, the spacer layer **5**, the first solid electrolyte layer **4**, the third substrate layer **3**, and the reference electrode **42** form an electrochemical sensor cell, namely, an oxygen-partial-pressure detection sensor cell **80** for main pump control.

(26) The oxygen concentration (oxygen partial pressure) in the first internal cavity **20** can be determined by measuring an electromotive force (voltage V_0) in the main-pump-control oxygen-partial-pressure detection sensor cell **80**. Furthermore, feedback control is performed on the pump voltage V_{p0} of a variable power source **24** such that the voltage V_0 becomes a target value, whereby the pump current I_{p0} is controlled. Accordingly, the oxygen concentration in the first internal cavity **20** can be maintained at a predetermined fixed value.

(27) The third diffusion control section **30** creates a predetermined diffusion resistance to the measurement-object gas whose oxygen concentration (oxygen partial pressure) has been controlled in the first internal cavity **20** by the operation of the main pump cell **21** and guides the measurement-object gas into the second internal cavity **40**.

(28) The second internal cavity **40** is provided as a space for further adjusting, using an auxiliary pump cell **50**, the oxygen concentration (oxygen partial pressure) of the measurement-object gas introduced through the third diffusion control section **30** after the oxygen partial pressure is adjusted in advance in the first internal cavity **20**. Thus, the oxygen concentration in the second internal cavity **40** can be maintained at a constant value with high accuracy so that the gas sensor **100** can measure the NO_x concentration with high accuracy.

(29) The auxiliary pump cell **50** is an auxiliary electrochemical pump cell composed of an auxiliary pump electrode **51** having a ceiling electrode portion **51a** disposed over substantially an entire portion of the lower surface of the second solid electrolyte layer **6** that faces the second internal cavity **40**, the outer pump electrode **23** (the outer electrode is not limited to the outer pump electrode **23**, but may be any suitable electrode outside the sensor element **101**), and the second solid electrolyte layer **6**.

(30) The auxiliary pump electrode **51** is disposed within the second internal cavity **40** in a tunnel-like structure similar to the aforementioned inner pump electrode **22** provided in the first internal cavity **20**. Specifically, the tunnel-like structure is formed such that the second solid electrolyte layer **6** that provides a ceiling surface for the second internal cavity **40** is provided with the ceiling electrode **51a**, the first solid electrolyte layer **4** that provides a bottom surface for the second internal cavity **40** is provided with a bottom electrode **51b**, and side electrodes (not shown) that connect the ceiling electrode **51a** and the bottom electrode **51b** are formed on opposite wall surfaces of the spacer layer **5** that provide sidewalls for the second internal cavity **40**. The auxiliary pump electrode **51** is similar to the inner pump electrode **22** in being formed by using a material with a weakened reduction ability against the NO.sub.x component in the measurement-object gas.

(31) In detail, the auxiliary pump electrode **51** contains a catalytically-active noble metal (e.g., at least one of Pt, Rh, Pd, Ru, and Ir). The auxiliary pump electrode **51** also contains a noble metal (e.g., Au) having the aforementioned catalytic-activity inhibition ability. The auxiliary pump electrode **51** is preferably formed of a cermet containing a noble metal and an oxygen-ion-conductive oxide (i.e., ZrO.sub.2). Moreover, the auxiliary pump electrode **51** is preferably porous.

In this embodiment, the auxiliary pump electrode **51** is a porous cermet electrode composed of Pt and ZrO.sub.2 and containing 1% Au.

(32) In the auxiliary pump cell **50**, the desired voltage V_{p1} is applied between the auxiliary pump electrode **51** and the outer pump electrode **23**. Thus, oxygen can be pumped from the atmosphere in the second internal cavity **40** to the external space or from the external space to the second internal cavity **40**.

(33) To control the oxygen partial pressure in the atmosphere in the second internal cavity **40**, the auxiliary pump electrode **51**, the reference electrode **42**, the second solid electrolyte layer **6**, the spacer layer **5**, the first solid electrolyte layer **4**, and the third substrate layer **3** form an electrochemical sensor cell, namely, an oxygen-partial-pressure detection sensor cell **81** for auxiliary pump control.

(34) The auxiliary pump cell **50** performs pumping in accordance with a variable power source **52** that is voltage-controlled based on an electromotive force (voltage V_1) detected by the auxiliary-pump-control oxygen-partial-pressure detection sensor cell **81**. Accordingly, the oxygen partial pressure in the atmosphere within the second internal cavity **40** is controlled to a low partial pressure that substantially has no effect on NO.sub.x measurement.

(35) In addition, a pump current I_{p1} is used for controlling the electromotive force of the main-pump-control oxygen-partial-pressure detection sensor cell **80**. In detail, the pump current I_{p1} is input as a control signal to the main-pump-control oxygen-partial-pressure detection sensor cell **80**, and the voltage V_0 thereof is controlled, whereby the gradient of the oxygen partial pressure in the measurement-object gas introduced to the second internal cavity **40** from the third diffusion control section **30** is controlled to be constantly fixed. In the case of application as a NO.sub.x sensor, the oxygen concentration within the second internal cavity **40** is maintained at a fixed value of about 0.001 ppm in accordance with the functions of the main pump cell **21** and the auxiliary pump cell **50**.

(36) The fourth diffusion control section **60** creates predetermined diffusion resistance to the measurement-object gas whose oxygen concentration (oxygen partial pressure) has been controlled in the second internal cavity **40** by the operation of the auxiliary pump cell **50** and guides the measurement-object gas into the third internal cavity **61**. The fourth diffusion control section **60** functions to limit the amount of NO_x flowing into the third internal cavity **61**.

(37) The third internal cavity **61** is provided as a space for processing associated with the measurement of the nitrogen oxide (NO_x) concentration in the measurement-object gas introduced through the fourth diffusion control section **60** after the oxygen concentration (oxygen partial pressure) is adjusted in advance in the second internal cavity **40**. NO_x concentration measurement is mainly performed in the third internal cavity **61** by the operation of a measurement pump cell **41**.

(38) The measurement pump cell **41** measures the NO.sub.x concentration in the measurement-object gas within the third internal cavity **61**. The measurement pump cell **41** is an electrochemical pump cell constituted of a measurement electrode **44** provided on the upper surface of the first solid electrolyte layer **4** facing the third internal cavity **61**, the outer pump electrode **23**, the second solid electrolyte layer **6**, the spacer layer **5**, and the first solid electrolyte layer **4**. The measurement electrode **44** is a porous cermet electrode composed of a material with an enhanced reduction ability against the NO.sub.x component in the measurement-object gas than the inner pump electrode **22**. The measurement electrode **44** also functions as a NO.sub.x reduction catalyst that reduces the NO.sub.x existing in the atmosphere within the third internal cavity **61**.

(39) In detail, the measurement electrode **44** contains a catalytically-active noble metal (e.g., at least one of Pt, Rh, Pd, Ru, and Ir). The content of a noble metal having the aforementioned catalytic-activity inhibition ability in the measurement electrode **44** is smaller than the content thereof in the main pump cell **21** and the auxiliary pump electrode **51**. The measurement electrode **44** preferably does not contain a noble metal having the catalytic-activity inhibition ability. The measurement electrode **44** is preferably formed of a cermet containing a noble metal and an

oxygen-ion-conductive oxide (i.e., ZrO.sub.2). Moreover, the measurement electrode **44** is preferably porous. In this embodiment, the measurement electrode **44** is a porous cermet electrode composed of Pt, Rh, and ZrO.sub.2 .

(40) The measurement pump cell **41** pumps out oxygen produced by the decomposition of nitrogen oxide in the atmosphere around the measurement electrode **44**. The amount of oxygen produced can be detected as a pump current I_{p2} .

(41) Furthermore, in order to detect the oxygen partial pressure around the measurement electrode **44**, the first solid electrolyte layer **4**, the third substrate layer **3**, the measurement electrode **44**, and the reference electrode **42** constitute an electrochemical sensor cell, that is, a measurement-pump-control oxygen-partial-pressure detection sensor cell **82**. A pulse power source **46** is controlled based on an electromotive force (voltage $V2$) detected by the measurement-pump-control oxygen-partial-pressure detection sensor cell **82**. The pulse power source **46** applies the pump current I_{p2} , which is repeatedly on-off controlled, between the measurement electrode **44** and the outer pump electrode **23**. The pulse power source **46** serves as an electric current source. The measurement pump cell **41** operates in accordance with this pump current I_{p2} .

(42) The measurement-object gas introduced to the second internal cavity **40** reaches the measurement electrode **44** in the third internal cavity **61** via the fourth diffusion control section **60** under a condition where the oxygen partial pressure is controlled. The nitrogen oxide in the measurement-object gas surrounding the measurement electrode **44** is reduced

($2\text{NO.fwdarw.N.sub.2}+\text{O.sub.2}$), so that oxygen is produced. Then, the produced oxygen is to undergo pumping by the measurement pump cell **41**. During the pumping of the oxygen, the pump current I_{p2} applied by the pulse power source **46** is controlled such that the voltage $V2$ detected by the measurement-pump-control oxygen-partial-pressure detection sensor cell **82** reaches a target state. Because the amount of oxygen produced around the measurement electrode **44** is proportional to the concentration of the nitrogen oxide in the measurement-object gas, the nitrogen oxide concentration in the measurement-object gas is calculated by using the pump current I_{p2} in the measurement pump cell **41**.

(43) An oxygen-partial-pressure detection device serving as an electrochemical sensor cell may be constituted by combining the measurement electrode **44**, the first solid electrolyte layer **4**, the third substrate layer **3**, and the reference electrode **42**. Thus, an electromotive force according to a difference between the amount of oxygen produced as a result of reduction of the NO_x component in the atmosphere surrounding the measurement electrode **44** and the amount of oxygen contained in reference air can be detected, so that the concentration of the NO_x component in the measurement-object gas can be determined accordingly.

(44) Furthermore, the second solid electrolyte layer **6**, the spacer layer **5**, the first solid electrolyte layer **4**, the third substrate layer **3**, the outer pump electrode **23**, and the reference electrode **42** constitute an electrochemical sensor cell **83**. The oxygen partial pressure in the measurement-object gas outside the sensor can be detected in accordance with an electromotive force (voltage V_{ref}) obtained by the sensor cell **83**.

(45) In the gas sensor **100** having the above configuration, the measurement pump cell **41** receives the measurement-object gas whose oxygen partial pressure is constantly maintained at a fixed low value (i.e., a value that substantially has no effect on NO.sub.x measurement) as a result of actuation of the main pump cell **21** and the auxiliary pump cell **50**. Thus, the NO.sub.x concentration in the measurement-object gas can be ascertained based on the pump current I_{p2} , used by the measurement pump cell **41** for pumping out oxygen produced by NO.sub.x reduction, substantially in proportion to the NO.sub.x concentration in the measurement-object gas.

(46) To increase the oxygen ion conductivity of the solid electrolyte, the sensor element **101** further includes a heater section **70** that functions as a temperature regulator to heat and maintain the temperature of the sensor element **101**. The heater section **70** includes a heater connector electrode **71**, a heater **72**, a through-hole **73**, a heater insulating layer **74**, and a pressure relief vent **75**.

(47) The heater connector electrode **71** is formed in contact with the lower surface of the first substrate layer **1**. The heater connector electrode **71** is connected to an external power supply so that the heater section **70** can be externally powered.

(48) The heater **72** is an electrical resistor formed between the second substrate layer **2** and the third substrate layer **3**. The heater **72** is connected to the heater connector electrode **71** through the through-hole **73**. The heater **72** is externally powered through the heater connector electrode **71** to generate heat, thereby heating and maintaining the temperature of the solid electrolyte forming the sensor element **101**.

(49) The heater **72** is embedded over the entire region from the first internal cavity **20** to the third internal cavity **61** so that the temperature of the entire sensor element **101** can be adjusted to a temperature that activates the solid electrolyte.

(50) The heater insulating layer **74** is an insulating layer covering the upper and lower surfaces of the heater **72** and formed of an insulator such as alumina. The heater insulating layer **74** is formed in order to ensure electrical insulation between the second substrate layer **2** and the heater **72** and electrical insulation between the third substrate layer **3** and the heater **72**.

(51) The pressure relief vent **75** extends through the third substrate layer **3** and the air introduction layer **48** so as to communicate with the reference gas introduction space **43**. The pressure relief vent **75** is formed in order to mitigate an increase in internal pressure due to a temperature increase in the heater insulating layer **74**.

(52) The controller **90** is a microprocessor including, for example, a CPU **92** and a memory **94**. The controller **90** receives the voltage **V0** detected by the main-pump-control oxygen-partial-pressure detection sensor cell **80**, the voltage **V1** detected by the auxiliary-pump-control oxygen-partial-pressure detection sensor cell **81**, the voltage **V2** detected by the measurement-pump-control oxygen-partial-pressure detection sensor cell **82**, the voltage **Vref** detected by the sensor cell **83**, the pump current **Ip0** detected by the main pump cell **21**, and the pump current **Ip1** detected by the auxiliary pump cell **50**. Furthermore, the controller **90** outputs control signals to the variable power source **24** of the main pump cell **21**, the variable power source **52** of the auxiliary pump cell **50**, and the pulse power source **46** of the measurement pump cell **41**, so as to control the cells **21**, **50**, and **41**.

(53) The controller **90** performs feedback control on the pump voltage **Vp0** of the variable power source **24** based on the voltage **V0** so as to set the voltage **V0** to a target value (referred to as "target value **V0***") (i.e., to set the oxygen concentration in the first internal cavity **20** to a target concentration). Thus, the pump current **Ip0** varies depending on the concentration of oxygen contained in the measurement-object gas.

(54) The controller **90** also performs feedback control on the voltage **Vp1** of the variable power source **52** based on the voltage **V1** so as to set the voltage **V1** to a target value (referred to as "target value **V1***") (i.e., to set the oxygen concentration in the second internal cavity **40** to a predetermined low oxygen concentration that substantially has no effect on NO.sub.x measurement). In addition, the controller **90** sets the target value **V0*** of the voltage **V0** (i.e., performs feedback control) based on the pump current **Ip1** so as to set the pump current **Ip1** flowing in accordance with the voltage **Vp1** to a target value (referred to as "target value **Ip1***"). Accordingly, the gradient of the oxygen partial pressure in the measurement-object gas introduced to the second internal cavity **40** from the third diffusion control section **30** is controlled to be constantly fixed. Moreover, the oxygen partial pressure in the atmosphere within the second internal cavity **40** is controlled to a low partial pressure that substantially has no effect on NO.sub.x measurement.

(55) Moreover, the controller **90** performs feedback control on the pump current **Ip2** of the pulse power source **46** based on the voltage **V2** so as to set the voltage **V2** to a target value (referred to as "target value **V2***") (i.e., to set the oxygen concentration in the third internal cavity **61** to a predetermined low concentration). Thus, oxygen produced as a result of reduction of the NOx in

the measurement-object gas in the third internal cavity **61** is pumped out from the third internal cavity **61** such that the oxygen becomes substantially zero. Then, the controller **90** determines the NO.sub.x concentration in the measurement-object gas based on the pump current I_{p2} serving as a value according to the oxygen produced in the third internal cavity **61** from the specific gas (i.e., NO_x).

(56) The memory **94** stores therein, for example, a relational expression (e.g., a linear function) or a map as a correspondence relationship between the pump current I_{p2} and the NO.sub.x concentration. Such a relational expression or a map can be preliminarily obtained from tests.

(57) An example of how the gas sensor **100** having the above-described configuration is used will be described below. It is assumed that the CPU **92** of the controller **90** is controlling the pump cells **21**, **41**, and **50** described above and is acquiring the voltages V_0 , V_1 , V_2 , and V_{ref} from the sensor cells **80** to **83** described above. In this state, when the measurement-object gas is introduced through the gas inlet **10**, the measurement-object gas first passes through the first diffusion control section **11**, the buffer space **12**, and the second diffusion control section **13** and reaches the first internal cavity **20**. Then, the oxygen concentration in the measurement-object gas is adjusted by the main pump cell **21** and the auxiliary pump cell **50** in the first internal cavity **20** and the second internal cavity **40**. After the adjustment, the measurement-object gas reaches the third internal cavity **61**. The CPU **92** then detects the NO_x concentration in the measurement-object gas based on the pump current I_{p2} and the correspondence relationship stored in the memory **94**.

(58) The operation of the measurement pump cell **41** and the control of the measurement pump cell **41** by the controller **90** will be described here in detail. FIG. **3** illustrates temporal changes in the pump current I_{p2} and the voltage V_2 . The upper section of FIG. **3** indicates the temporal change in the pump current I_{p2} , and the lower section indicates the temporal change in the voltage V_2 . With regard to the pump current I_{p2} , the direction in which oxygen can be pumped out from around the measurement electrode **44** to around the outer pump electrode **23** is defined as “positive”. In FIG. **3**, the upward direction of the ordinate axis is defined as “positive direction”. The positive direction of the pump current I_{p2} is the direction of an arrow of the pump current I_{p2} in FIG. **1**, and is the direction in which electric current flows from the measurement electrode **44** toward the outer pump electrode **23** at the outer side of the sensor element **101**. With regard to the voltage V_2 , a state where the electric potential of the reference electrode **42** is higher than that of the measurement electrode **44** is defined as “positive”. In FIG. **3**, the upward direction of the ordinate axis is defined as “positive direction”.

(59) The pulse power source **46** of the measurement pump cell **41** applies the repeatedly on-off controlled pump current I_{p2} , that is, the intermittent pump current I_{p2} , between the measurement electrode **44** and the outer pump electrode **23**, so that oxygen is pumped out from around the measurement electrode **44**. In this embodiment, as shown in FIG. **3**, the pump current I_{p2} is a pulse-wave electric current that is repeatedly on-off controlled in cycles T . For example, when the pump current I_{p2} rises from 0 A at a time point t_1 and transitions to an on mode, the pump current I_{p2} becomes a maximum current I_{p2max} and maintains this state until an ON time T_{on} elapses and a time point t_4 is reached. When the pump current I_{p2} falls at the time point t_4 and transitions to an off mode, the pump current I_{p2} becomes 0 A until an OFF time T_{off} elapses and a time point t_7 is reached. Although the actual pump current I_{p2} requires a short period of time for rising from the time point t_1 and for falling from the time point t_4 , this is omitted in FIG. **3**. Moreover, there may be a case where electric current slightly flows due to the effect of, for example, noise even while the pump current I_{p2} output from the pulse power source **46** is in the off mode. However, such a case is omitted in FIG. **3**.

(60) As mentioned above, the CPU **92** of the controller **90** performs feedback control on the pump current I_{p2} of the pulse power source **46** based on the voltage V_2 so as to set the oxygen concentration in the third internal cavity **61** to the predetermined low concentration. An amount of oxygen that the measurement pump cell **41** pumps out from the third internal cavity **61** during one

cycle (i.e., cycle T) in accordance with the pump current I_{p2} is proportional to an average value I_{p2ave} (see a single-dot chain line in the upper section of FIG. 3) of the pump current I_{p2} during one cycle. Therefore, for example, the CPU 92 varies the average value I_{p2ave} by outputting a control value to the pulse power source 46 to change at least one of parameters including the proportion (i.e., duty ratio) of the ON time T_{on} occupying the cycle T, the cycle T, and the maximum current I_{p2max} . Alternatively, the CPU 92 may output the average value I_{p2ave} as a control value to the pulse power source 46, and the pulse power source 46 may vary at least one of the aforementioned parameters based on the control value. In this embodiment, the CPU 92 outputs the duty ratio (or a variation of the duty ratio) as a control value to the pulse power source 46, and the pulse power source 46 varies the duty ratio of the pump current I_{p2} based on the control value. For example, if the voltage $V2$ is smaller than the target value $V2^*$, the CPU 92 controls the pulse power source 46 to increase the duty ratio of the pump current I_{p2} (i.e., to extend the ON time T_{on} without changing the cycle T), thereby reducing the oxygen concentration in the third internal cavity 61. FIG. 3 shows an example where the second ON time T_{on} (between time points $t7$ and $t8$) is twice as long as the first ON time T_{on} (between the time points $t1$ and $t4$), such that the average value I_{p2ave} in the second cycle T is twice as large as that in the first cycle T. FIG. 3 also shows that, with the average value I_{p2ave} being doubled, the oxygen concentration in the third internal cavity 61 is reduced, such that the voltage $V2$ is higher with a value $V2b'$ at a time point $t9$ than a value $V2b$ at the time point $t7$.

(61) The pump current I_{p2} controlled by the CPU 92 in this manner ultimately becomes an electric current according to the oxygen produced in the third internal cavity 61 from the specific gas (i.e., NO_x). Therefore, the CPU 92 can determine the NO_x concentration in the measurement-object gas based on this pump current I_{p2} . For example, a value of a parameter (i.e., duty ratio) of the pump current I_{p2} used as a control value by the CPU 92 is proportional to the NO_x concentration in the measurement-object gas. Therefore, by storing the correspondence relationship between the two in the memory 94, the CPU 92 can determine the NO_x concentration based on the control value output to the pulse power source 46 by the CPU 92 and the correspondence relationship stored in the memory 94. Likewise, the CPU 92 can also determine the NO_x concentration by using a value (e.g., average value I_{p2ave}) determined based on the control value.

(62) The following description relates to a comparative example where the pump current I_{p2} is a continuous electric current. FIG. 4 is a schematic cross-sectional view of a gas sensor 900 according to a comparative example. FIG. 5 illustrates a temporal change in the pump current I_{p2} of the gas sensor 900. Components of the gas sensor 900 that are identical to those of the gas sensor 100 are given the same reference signs as those in FIG. 1, and detailed descriptions thereof will be omitted. The definition of the direction of the pump current I_{p2} is the same as that of the gas sensor 100, and the upward direction of the ordinate axis in FIG. 5 is defined as “positive direction”. Furthermore, in FIG. 5, the waveform of the pump current I_{p2} shown in FIG. 3 is indicated by a single-dot chain line as a comparison. The gas sensor 900 includes a measurement pump cell 941 in place of the measurement pump cell 41. The measurement pump cell 941 is similar to the measurement pump cell 41 except that the measurement pump cell 941 includes a variable power source 946 in place of the pulse power source 46. The variable power source 946 applies a pump voltage V_{p2} between the measurement electrode 44 and the outer pump electrode 23. Accordingly, the measurement pump cell 941 receives a continuous pump current I_{p2} , as shown in FIG. 5. A controller of the gas sensor 900 is not shown, but is similar to the controller 90 in that the controller receives the voltages $V0$, $V1$, $V2$, and V_{ref} and the pump currents I_{p0} and I_{p1} and outputs control signals to the variable power sources 24 and 52. Furthermore, the controller of the gas sensor 900 performs feedback control on the pump voltage V_{p2} of the variable power source 946 based on the input voltage $V2$ so as to set the voltage $V2$ to a target value. Thus, oxygen produced as a result of reduction of the NO_x in the measurement-object gas in the third internal cavity 61 is pumped out from the third internal cavity 61 by the measurement pump cell 941 such

that the oxygen becomes substantially zero. For example, when the controller is to perform feedback control on the pump voltage V_{p2} at every cycle T identical to that in FIG. 3, the pump current I_{p2} flowing to the measurement pump cell **941** changes at every cycle T , as shown in FIG. 5, while flowing continuously. Although the actual pump current I_{p2} requires a short period of time to completely change at every cycle T , this is omitted in FIG. 5. The controller of the gas sensor **900** then acquires the pump current I_{p2} from the measurement pump cell **941** and calculates the NO_x concentration in the measurement-object gas based on this pump current I_{p2} .

(63) If the average value I_{p2ave} in FIG. 3 and the pump current I_{p2} indicated by a solid line in FIG. 5 are the same, the amount of oxygen pumped out from the third internal cavity **61** at every cycle T is the same. Therefore, even in a case where the pump current I_{p2} is applied intermittently as in this embodiment instead of the pump current I_{p2} being applied continuously as in the comparative example, the CPU **92** can still adjust the oxygen concentration in the third internal cavity **61** and measure the NO_x concentration. Moreover, because the gas sensor **100** uses the pulse power source **46** serving as an electric current source, the control value output to the pulse power source **46** by the CPU **92** is a value according to the pump current I_{p2} . Therefore, without having to receive the pump current I_{p2} from the measurement pump cell **41**, the CPU **92** can ascertain the pump current I_{p2} based on the control value set by the CPU **92** and determine the NO_x concentration.

(64) In the gas sensor **100**, the pump current I_{p2} is intermittently applied to the measurement pump cell **41** so that a change in the catalytic activity of the measurement electrode **44** due to use of the gas sensor **100** can be suppressed, as compared with a case where the pump current I_{p2} is continuously applied to the measurement pump cell **941**, as in the gas sensor **900**. A conceivable reason for this will be described below.

(65) When oxygen is to be pumped out from around the measurement electrode **44** by applying the pump current I_{p2} between the measurement electrode **44** and the outer pump electrode **23**, the surrounding oxygen (i.e., oxygen mainly produced as a result of NO_x reduction) turns into oxygen ions (O_{2+4e^-}) inside the measurement electrode **44**. The oxygen ions serve as electron carriers and travel through the solid electrolyte layers (i.e., the layers **4** to **6**) toward the outer pump electrode **23**. During this process, if the pump current I_{p2} is a continuous electric current, a reaction where some noble metals (i.e., Pt and Rh) in the measurement electrode **44** oxidize and a reaction where oxygen ions are released as a result of reduction of the oxidized noble metals (i.e., PtO, PtO₂, and Rh₂O₃) both occur. When both reactions reach a state of equilibrium, some of the noble metals in the measurement electrode **44** are in a constantly oxidized state. Since oxidized noble metals tend to evaporate more easily than before they are oxidized, the noble metals in the measurement electrode **44** tend to decrease with use of the gas sensor **900**, thus causing the catalytic activity of the measurement electrode **44** to decrease. Moreover, the oxidization of the noble metals in the measurement electrode **44** may conceivably cause the microstructure of the measurement electrode **44** to change. This may conceivably cause the three-phase interface among the measurement electrode **44**, the pores in the measurement electrode **44**, and the first solid electrolyte layer **4** or the two-phase interface between the measurement electrode **44** and the first solid electrolyte layer **4** to decrease in surface area, thus causing the catalytic activity of the measurement electrode **44** to decrease.

(66) On the other hand, the following description relates to a case where an intermittent pump current I_{p2} is used in the gas sensor **100** and is intermittently applied to the measurement pump cell **41** such that the intermittent pump current I_{p2} becomes an average electric current (i.e., average value I_{p2ave}) equal to the continuous pump current I_{p2} in the gas sensor **900**. In this case, the pump current I_{p2} (i.e., maximum current I_{p2max}) applied to the measurement electrode **44** during an on mode (e.g., between the time points $t1$ and $t4$ in FIG. 3) in the gas sensor **100** is a value larger than the continuous pump current I_{p2} in the gas sensor **900** (see FIGS. 3 and 5). Accordingly, the oxygen inside the measurement electrode **44** turns into ions and travel toward the outer pump

electrode **23** more during the on mode of the intermittent pump current I_{p2} than when the pump current I_{p2} is applied continuously, so that the oxygen concentration inside the measurement electrode **44** decreases. In such a state where the oxygen concentration inside the measurement electrode **44** is low, the aforementioned noble metals in the measurement electrode **44** are less likely to oxidize, and rather, reduction of oxidized noble metals may occur. Thus, in the gas sensor **100**, a decrease in the aforementioned noble metals in the measurement electrode **44** with use thereof is suppressed, as compared with the gas sensor **900**. Furthermore, in the gas sensor **100**, electric current hardly flows to the measurement electrode **44** when the pump current I_{p2} is in an off mode (e.g., between the time points t_4 and t_7 in FIG. 3), so that oxidization of the aforementioned noble metals in the measurement electrode **44** is less likely to occur. As a result, in the gas sensor **100**, oxidization of the noble metals in the measurement electrode **44** is suppressed in both an on mode and an off mode of the pump current I_{p2} . It is thus conceivable that a decrease in the catalytic activity of the measurement electrode **44** due to use of the gas sensor **100** is suppressed therein, as compared with the gas sensor **900**. Since a lower catalytic activity of the measurement electrode **44** leads to suppressed NO_x reduction in the third internal cavity **61**, the pump current I_{p2} (more specifically, the average value I_{p2ave}) decreases, thus resulting deteriorated sensitivity for detecting the NO_x concentration. In the gas sensor **100** according to this embodiment, such deteriorated sensitivity for detecting the NO_x concentration with use thereof can be suppressed. Therefore, the gas sensor **100** according to this embodiment can maintain the measurement accuracy over a long period of use, and can thus extend the lifespan thereof.

(67) The voltage V_2 is a voltage between the measurement electrode **44** and the reference electrode **42** and is basically a value according to the oxygen concentration in the third internal cavity **61**. However, because the electric potential of the measurement electrode **44** changes intermittently when the pulse power source **46** intermittently applies the pump current I_{p2} thereto, the voltage V_2 also varies periodically in a pulsating manner in response to this effect. In detail, the waveform of the voltage V_2 has a first period in which a change is occurring as a result of the pump current I_{p2} being controlled to an on mode, and a second period in which the change has receded as a result of the pump current I_{p2} being controlled to an off mode. For example, in FIG. 3, with regard to the voltage V_2 , a change begins (i.e., the voltage V_2 begins to rise) from the time point t_1 , a value V_{2a} (i.e., maximum value) corresponding to a maximum change is reached at the time point t_4 , the change begins to recede (i.e., the voltage V_2 begins to fall) from the time point t_4 , and a value V_{2b} (i.e., minimum value) where the change has receded the most is reached at the time point t_7 . In this case, the range from the value V_{2b} to the value V_{2a} of the voltage V_2 during the duration of a single cycle T of the pump current I_{p2} is defined as 0% to 100%, and the first period and the second period of the voltage V_2 are set using this range as a reference. In detail, a period in which the voltage V_2 is 90% or higher (i.e. between the time points t_3 and t_5) is defined as the first period, and the length thereof is defined as a first duration T_1 . A period from when the pump current I_{p2} is controlled to an off mode and the voltage V_2 becomes 10% or lower to when the voltage V_2 begins to rise as a result of the pump current I_{p2} being controlled to an on mode in the subsequent cycle (i.e. between the time points t_6 and t_7) is defined as a second period, and the length thereof is defined as a second duration T_2 . In FIG. 3, the voltage V_2 reaches the value V_{2a} corresponding to the first maximum change at the time point t_4 at which the pump current I_{p2} is controlled to an off mode. However, if the ON time T_{on} is long, there may be a case where the voltage V_2 may reach the value V_{2a} prior to the time point t_4 and remain in this state until the time point t_4 .

(68) Accordingly, in this embodiment, the voltage V_2 varies periodically in response to the effect of the pump current I_{p2} . During the second period, the effect that the pump current I_{p2} has on the voltage V_2 is smaller, as compared with the first period. Therefore, the voltage V_2 during the second period is a value that indicates the oxygen concentration in the third internal cavity **61** more accurately than the voltage V_2 during the first period. Accordingly, the CPU **92** preferably

performs the above-described control of the pump current I_{p2} based on the voltage $V2$ during the second period. Thus, the CPU **92** can accurately set the oxygen concentration in the third internal cavity **61** close to the predetermined low concentration, and can also accurately determine the NO_x concentration in the measurement-object gas. In this embodiment, the CPU **92** acquires the voltage $V2$ during the second period, compares the acquired voltage $V2$ with the target value $V2^*$, and sets the control value to be output to the pulse power source **46**. Furthermore, as shown in FIG. 3, the voltage $V2$ during the second period also changes with time, and the effect of the pump current I_{p2} tends to decrease toward the end of the second period (i.e., the time point $t7$ in FIG. 3). This is conceivably due to the effect of a capacity component in the measurement pump cell **41**, such as a capacity component of the measurement electrode **44**. Thus, the CPU **92** preferably controls the pump current I_{p2} based on the voltage $V2$ at the latest possible timing in the second period. For example, the CPU **92** may control the pump current I_{p2} based on the voltage $V2$ at any timing in the latter half of the second period.

(69) The CPU **92** may control the pump current I_{p2} based on the voltage $V2$ during the second period in each of a plurality of cycles T (e.g., based on an average value of a plurality of voltages $V2$ acquired in different cycles). For example, if the pump current I_{p2} is controlled based only on the voltage $V2$ during the second period in one cycle, the control of the pump current I_{p2} may sometimes become unstable due to, for example, a fluctuation or a deviation of the actual average value I_{p2ave} relative to the proper average value I_{p2ave} corresponding to the NO_x concentration. In contrast, the CPU **92** controls the pump current I_{p2} based on, for example, the average value of the voltages $V2$ during the second periods in the plurality of cycles T , so that the control is less likely to become unstable. In this embodiment, the CPU **92** stores the voltage $V2$ in the memory **94** at a predetermined timing during the second period for every cycle T , and controls the pump current I_{p2} based on an average value of multiple times' (e.g., three times') worth of most-recently-stored voltages $V2$.

(70) The CPU **92** may control the main pump cell **21** based on the voltage $V0$ during the second period of the voltage $V2$. Likewise, the CPU **92** may control the auxiliary pump cell **50** based on the voltage $V1$ during the second period of the voltage $V2$. The CPU **92** may set the target value $V0^*$ based on the pump current I_{p1} during the second period of the voltage $V2$.

(71) The correspondence relationship between the components in this embodiment and the components in the present invention will now be clarified. The first substrate layer **1**, the second substrate layer **2**, the third substrate layer **3**, the first solid electrolyte layer **4**, the spacer layer **5**, and the second solid electrolyte layer **6** according to this embodiment correspond to an element body according to the present invention. The main pump cell **21**, the auxiliary pump cell **50**, and the measurement pump cell **41** each correspond to a pump cell. The inner pump electrode **22**, the auxiliary pump electrode **51**, and the measurement electrode **44** each correspond to an inner electrode. The outer pump electrode **23** corresponds to an outer electrode. Furthermore, the inner pump electrode **22** corresponds to an inner main pump electrode, the auxiliary pump electrode **51** corresponds to an inner auxiliary pump electrode, the measurement electrode **44** corresponds to an inner measurement electrode, the outer pump electrode **23** corresponds to an outer main pump electrode, an outer auxiliary pump electrode, and an outer measurement electrode, and the third internal cavity **61** corresponds to a measurement chamber. Moreover, the pump current I_{p2} corresponds to a measurement pump current, the voltage $V2$ corresponds to a measurement voltage, the measurement-pump-control oxygen-partial-pressure detection sensor cell **82** corresponds to a measurement-voltage detection device, and the controller **90** corresponds to a measurement-pump-cell control device and a specific-gas-concentration detection device.

(72) The gas sensor **100** according to this embodiment described in detail above includes the main pump cell **21**, the auxiliary pump cell **50**, and the measurement pump cell **41** as one or more pump cells. The main pump cell **21** has the inner pump electrode **22** disposed in the first internal cavity **20** and also has the outer pump electrode **23**, and pumps out oxygen from the first internal cavity

20. The auxiliary pump cell 50 has the auxiliary pump electrode 51 disposed in the second internal cavity 40 at the downstream side of the first internal cavity 20 and also has the outer pump electrode 23, and pumps out oxygen from the second internal cavity 40. The measurement pump cell 41 has the measurement electrode 44 disposed in the third internal cavity 61 at the downstream side of the second internal cavity 40 and also has the outer pump electrode 23, and pumps out oxygen produced in the third internal cavity 61 from NO_x in the measurement-object gas. Of these pump cells, the measurement pump cell 41 applies the repeatedly on-off controlled pump current Ip2 between the measurement electrode 44 and the outer pump electrode 23, so as to pump out oxygen surrounding the measurement electrode 44. Accordingly, the gas sensor 100 can suppress a change (i.e., decrease) in the catalytic activity of the measurement electrode 44 with use thereof, as compared with a case where the pump current Ip2 is continuously applied between the measurement electrode 44 and the outer pump electrode 23, as in the gas sensor 900. The repeatedly on-off controlled pump current Ip2 has a higher peak value (i.e., maximum current Ip2max) than the continuous pump current Ip2, and thus achieves a high signal-to-noise ratio (S/N ratio) and is resistant to noise.

(73) Of the pump cells 21, 50, and 41, the measurement pump cell 41 operates by using an intermittent pump current (i.e., pump current Ip2). A change in the catalytic activity of the measurement electrode 44 has a larger effect on the sensitivity for detecting the NO_x concentration of the gas sensor 100 than a change in the catalytic activity of the inner pump electrode 22 and the auxiliary pump electrode 51. Therefore, deterioration in the detection sensitivity may be readily minimized by suppressing a change in the catalytic activity of the measurement electrode 44.

(74) Furthermore, the CPU 92 controls the pump current Ip2 based on the voltage V2 during the second period so as to set the oxygen concentration in the third internal cavity 61 to the predetermined low concentration. Then, the CPU 92 detects the NO_x concentration in the measurement-object gas based on the pump current Ip2. Accordingly, the CPU 92 can accurately detect the NO_x concentration.

(75) The present invention is not limited whatsoever to the above embodiment, and various embodiments are possible so long as they belong within the technical scope of the present invention.

(76) For example, although the intermittent pump current Ip2 is an electric current having one rectangular wave per cycle (i.e., a rectangular single pulse current) in the above-described embodiment, as shown in FIG. 3, the present invention is not limited thereto. For example, the pulse power source 46 may use a burst pulse current, as shown in FIG. 6, as the intermittent pump current Ip2. In this case, the CPU 92 may vary the average value Ip2ave of the pump current Ip2 during one cycle by outputting a control value to the pulse power source 46 to change at least one of parameters including the proportion (i.e., duty ratio) of an oscillation period TA occupying a cycle T (i.e., burst cycle), the cycle T, the number of pulses during one cycle (i.e., five pulses in FIG. 6), a time Ta of a single oscillation (i.e., pulse), a pulse cycle (Ta+Tb), and the maximum current Ip2max.

(77) In the case where the pump current Ip2 is a burst pulse current, as in FIG. 6, the first period of the voltage V2 is included in the oscillation period TA, and the second period is included in a non-oscillation period TB. More specifically, in the case where the pump current Ip2 is a burst pulse current, the oscillation period TA is regarded as an on period (i.e., between the time points t1 and t4 in FIG. 3) of the pump current Ip2, the non-oscillation period TB is regarded as an off period (i.e., between the time points t4 and t7 in FIG. 3) of the pump current Ip2, and the second period is defined based on a method similar to that in the above-described embodiment.

(78) Although the pulse power source 46 applies a rectangular-wave pulse current as the pump current Ip2 in the above-described embodiment, as shown in FIG. 3, the pulse current to be applied is not limited to a rectangular-wave (square-wave) pulse current and may have a sinusoidal half waveform, a triangular waveform, a saw-tooth waveform, or a waveform at the time of discharge,

or may have a waveform obtained by combining one or more of these waveforms.

(79) Although not specifically described in the above-described embodiment, each cycle **T** of the intermittent pump current **Ip2** in FIGS. **3** and **6** may be, for example, 0.1 s or shorter (i.e., a frequency of 10 Hz or higher), 0.02 s or shorter (i.e., a frequency of 50 Hz or higher), or 0.001 s or shorter (i.e., a frequency of 100 Hz or higher). For example, in a case where the controller **90** outputs the determined NO_x concentration to another device, such as an engine ECU of a vehicle, at every predetermined cycle **T_{out}**, the cycle **T** is preferably shorter than or equal to 1/10 of the cycle **T_{out}**.

(80) Although the measurement pump cell **41** among the pump cells **21**, **50**, and **41** included in the gas sensor **100** operates by using an intermittent pump current (i.e., pump current **Ip2**) in the above-described embodiment, the present invention is not limited thereto. At least one of the pump cells included in the gas sensor **100** may simply operate by using an intermittent pump current. For example, the main pump cell **21** and the measurement pump cell **41** may operate by using an intermittent pump current. FIG. **7** is a schematic cross-sectional view of a gas sensor **200** according to a modification. Components of the gas sensor **200** that are identical to those of the gas sensor **100** are given the same reference signs as those in FIG. **1**, and detailed descriptions thereof will be omitted. The gas sensor **200** includes a main pump cell **221** in place of the main pump cell **21**, and includes an auxiliary pump cell **250** in place of the auxiliary pump cell **50**. The main pump cell **221** is similar to the main pump cell **21** except that the main pump cell **221** includes a pulse power source **224**, serving as an electric current source similar to the pulse power source **46**, in place of the variable power source **24**. The auxiliary pump cell **250** is similar to the auxiliary pump cell **50** except that the auxiliary pump cell **250** includes a pulse power source **252**, serving as an electric current source similar to the pulse power source **46**, in place of the variable power source **52**. In the gas sensor **200** according to this modification, the main pump cell **221** applies an intermittent pulse current so that a change in the catalytic activity of the inner pump electrode **22** due to use of the gas sensor **200** can be suppressed. Moreover, the pulse power source **252** applies an intermittent pulse current so that a change in the catalytic activity of the auxiliary pump electrode **51** due to use of the gas sensor **200** can be suppressed. As mentioned above, the inner pump electrode **22** contains a catalytically-active noble metal and a noble metal having the catalytic-activity inhibition ability. Therefore, when a noble metal in the inner pump electrode **22** oxidizes and evaporates, it is conceivable that the oxygen pumping performance of the main pump cell **21** may deteriorate due to a decrease in the catalytic activity of the inner pump electrode **22**, or the inner pump electrode **22** may decompose the NO_x in the first internal cavity **20** due to an increase in the catalytic activity. Since either case is not preferable for the gas sensor **200**, it is significant to suppress such a change (i.e., an increase or decrease) in the catalytic activity of the inner pump electrode **22**. The same applies to the auxiliary pump electrode **51**. In a case where the main pump cell **221** is to pump oxygen into the first internal cavity **20** from around the outer pump electrode **23** by using an intermittent pulse current, it is conceivable that a change in the catalytic activity of the outer pump electrode **23** can be suppressed. For example, in a case where the oxygen concentration in the measurement-object gas is lower than the aforementioned target oxygen concentration in the first internal cavity **20** (including a case of a rich atmosphere), the controller **90** controls the pulse power source **224** such that the main pump cell **221** pumps oxygen into the first internal cavity **20** from around the outer pump electrode **23**. In this case, the controller **90** may invert the positive and negative signs of pulses of the pulse power source **224**, so as to switch between a mode where the main pump cell **221** pumps oxygen out from the first internal cavity **20** and a mode where the main pump cell **221** pumps oxygen into the first internal cavity **20**. Accordingly, a negative pulse current (i.e., a pulse whose electric current value falls from zero to a negative value) may be applied, instead of a positive pulse current as in FIGS. **3** and **6**.

(81) Even in a case where a continuous pump current **Ip0** is applied to the main pump cell **21** as in the above-described embodiment, an electric current source may be used in place of the variable

power source **24**, and the pump current I_{p0} may be varied directly by the electric current source. Likewise, even in a case where a continuous pump current I_{p1} is applied to the auxiliary pump cell **50** as in the above-described embodiment, an electric current source may be used in place of the variable power source **52**, and the pump current I_{p1} may be varied directly by the electric current source.

(82) Although the sensor element **101** of the gas sensor **100** includes the first internal cavity **20**, the second internal cavity **40**, and the third internal cavity **61** in the above-described embodiment, the present invention is not limited thereto. For example, the third internal cavity **61** may be omitted, as in a sensor element **301** in FIG. **8**. In the sensor element **301** according to a modification shown in FIG. **8**, the gas inlet **10**, the first diffusion control section **11**, the buffer space **12**, the second diffusion control section **13**, the first internal cavity **20**, the third diffusion control section **30**, and the second internal cavity **40** are provided next to each other in a communicating manner in that order between the lower surface of the second solid electrolyte layer **6** and the upper surface of the first solid electrolyte layer **4**. The measurement electrode **44** is disposed on the upper surface of the first solid electrolyte layer **4** within the second internal cavity **40**. The measurement electrode **44** is covered by a fourth diffusion control section **45**. The fourth diffusion control section **45** is a film formed of a ceramic porous body composed of, for example, alumina (Al.sub.2O.sub.3). Similar to the fourth diffusion control section **60** according to the above-described embodiment, the fourth diffusion control section **45** has a function of limiting the amount of NO_x flowing to the measurement electrode **44**. Moreover, the fourth diffusion control section **45** also functions as a protective film for the measurement electrode **44**. The ceiling electrode **51a** of the auxiliary pump electrode **51** is provided to extend to a position directly above the measurement electrode **44**. The sensor element **301** having such a configuration is similar to that in the above-described embodiment in being able to detect the NO_x concentration based on the pump current I_{p2} . In this case, the space surrounding the measurement electrode **44** (i.e., the interior of the fourth diffusion control section **45**) functions as a measurement chamber.

(83) Although the outer pump electrode **23** serves as an outer main pump electrode of the main pump cell **21**, an outer auxiliary pump electrode of the auxiliary pump cell **50**, and an outer measurement electrode of the measurement pump cell **41** in the above-described embodiment, the present invention is not limited thereto. Any one or two of the outer main pump electrode, the outer auxiliary pump electrode, and the outer measurement electrode may be provided at the outer side of the sensor element **101** independently of the outer pump electrode **23**.

(84) Although the element body of the sensor element **101** according to the embodiment described above is a stack including a plurality of solid electrolyte layers (the layers **1** to **6**), the present invention is not limited thereto. The element body of the sensor element **101** may include at least one oxygen-ion-conductive solid electrolyte layer and have a measurement-object gas flow section inside the element body. For example, the layers **1** to **5** other than the second solid electrolyte layer **6** in FIG. **1** may be layers formed of materials other than solid electrolytes (e.g., alumina layers). In this case, the electrodes of the sensor element **101** may be disposed on the second solid electrolyte layer **6**. For example, the measurement electrode **44** in FIG. **1** may be disposed on the lower surface of the second solid electrolyte layer **6**. In addition, the reference gas introduction space **43** may be disposed in the spacer layer **5** rather than in the first solid electrolyte layer **4**. The air introduction layer **48** may be disposed between the second solid electrolyte layer **6** and the spacer layer **5** rather than between the first solid electrolyte layer **4** and the third substrate layer **3**. The reference electrode **42** may be disposed on the lower surface of the second solid electrolyte layer **6** on the rear side of the third internal cavity **61**.

(85) Although the gas sensor **100** detects the NO_x concentration as a specific gas concentration in the above-described embodiment, the specific gas concentration may alternatively be the concentration of another oxide. In a case where the specific gas is an oxide, oxygen is produced as a result of reduction of the specific gas itself in the third internal cavity **61**, similarly to the above-

described embodiment. Thus, the CPU **92** can detect the specific gas concentration based on a detection value (e.g., average value I_{p2ave}) according to this oxygen. Alternatively, the specific gas may be a non-oxide, such as ammonia. In a case where the specific gas is a non-oxide, the specific gas is converted into an oxide in, for example, the first internal cavity **20** (i.e., is converted into NO by being oxidized in the case of ammonia), so that oxygen is produced as a result of reduction of the oxide in the third internal cavity **61** after the conversion. Thus, the CPU **92** can acquire a detection value according to this oxygen and detect the specific gas concentration. Accordingly, whether the specific gas is an oxide or a non-oxide, the gas sensor **100** can detect the specific gas concentration based on oxygen produced in the third internal cavity **61** from the specific gas.

(86) As an alternative to the above-described embodiment in which the controller **90** sets the target value $V0^*$ of the voltage $V0$ (i.e., performs feedback control) based on the pump current I_{p1} so as to set the pump current I_{p1} to the target value I_{p1}^* , and performs feedback control on the pump voltage $Vp0$ so as to set the voltage $V0$ to the target value $V0^*$, the controller **90** may perform another control. For example, the controller **90** may perform feedback control on the pump voltage $Vp0$ based on the pump current I_{p1} so as to set the pump current I_{p1} to the target value I_{p1}^* .

(87) Specifically, the controller **90** may omit the acquisition of the voltage $V0$ from the main-pump-control oxygen-partial-pressure detection sensor cell **80** and the setting of the target value $V0^*$ and may directly control the pump voltage $Vp0$ (and, by extension, the pump current I_{p0}) based on the pump current I_{p1} .

(88) Although the pump voltage $Vp2$ of the pulse power source **46** is controlled such that the voltage $V2$ detected by the measurement-pump-control oxygen-partial-pressure detection sensor cell **82** becomes the target value $V2^*$, and the NO_x concentration in the measurement-object gas is calculated by using the pump current I_{p2} at that time in the above-described embodiment, the present invention is not limited thereto. For example, the CPU **92** may perform feedback control on the measurement pump cell **41** (e.g., control the pump voltage $Vp2$) so as to set the average value I_{p2ave} shown in FIG. **3** to a predetermined target value I_{p2}^* , and may calculate the NO_x concentration by using the voltage $V2$ at that time. With the measurement pump cell **41** being controlled such that the average value I_{p2ave} becomes the target value I_{p2}^* , oxygen is pumped out from the third internal cavity **61** at a substantially fixed flow rate on average, or in other words, in terms of a longer time period than the cycle T . Therefore, the oxygen concentration in the third internal cavity **61** changes in accordance with the amount of oxygen produced as a result of reduction of the NO_x in the measurement-object gas in the third internal cavity **61**, thus causing the voltage $V2$ to change accordingly. Accordingly, the voltage $V2$ becomes a value according to the NO_x concentration in the measurement-object gas. Thus, the NO_x concentration can be calculated based on this voltage $V2$. For example, the correspondence relationship between the voltage $V2$ and the NO_x concentration may be preliminarily stored in the memory **94**. Moreover, the CPU **92** uses the value of the voltage $V2$ during the second period as a value of the voltage $V2$ for calculating the NO_x concentration. As mentioned above, the effect that the pump current I_{p2} has on the voltage $V2$ is small during the second period. Therefore, the specific gas concentration in the measurement-object gas can be accurately detected by using the voltage $V2$ during the second period.

(89) In FIGS. **3** and **6** in the above-described embodiment, the electric current waveform is rectangular in a case where the pump current I_{p1} is a pulse current. However, as mentioned above, a pulse current requires a short period of time for rising and falling. Specifically, a pulse current actually has a rise time and a fall time, and the waveform of a pulse current is not a perfect rectangle. Therefore, for example, as shown in an enlarged view of a pulse current in the upper section of FIG. **9**, if the pulse width of the pulse current is too small, the actual peak value of the pulse current (i.e., the peak value of the waveform indicated by a dashed line) may sometimes not reach the peak value of an ideal rectangular waveform (i.e., the peak value of the waveform indicated by a solid line) due to the effect of the rise time. In this case, for example, even if the

CPU **92** performs the aforementioned feedback control based on the voltage V_2 and the target value V_2^* and outputs a control value to the pulse power source **46** so as to set the average value I_{p2ave} to a certain target value (referred to as “target average current I_{p2ave}^* ”), the actual average value I_{p2ave} becomes smaller than the theoretical average value I_{p2ave} , thus causing the actual average value I_{p2ave} to deviate from the target average current I_{p2ave}^* . Thus, the controller **90** may sometimes be not able to properly control the measurement pump cell **41**, or the S/N ratio of the pump current I_{p2} may sometimes decrease. In contrast, for example, as shown in an enlarged view of a pulse current in the lower section of FIG. **9**, if the pulse width of the pulse current is large, the actual peak value of the pulse current (i.e., the peak value of the waveform indicated by a dashed line) reaches the peak value of an ideal rectangular waveform (i.e., the peak value of the waveform indicated by a solid line) even with the existence of the rise time. Therefore, if the pulse width is large, the actual average value I_{p2ave} is less likely to deviate from the theoretical average value I_{p2ave} and the target average current I_{p2ave}^* . Thus, the controller **90** preferably controls the measurement pump cell **41** such that the pulse width of the pump current I_{p2} does not become too small to an extent that the actual peak value does not reach the peak value of the ideal rectangular waveform. In other words, the controller **90** preferably controls the pump current I_{p2} in a range where the pulse width of the pump current I_{p2} does not fall below a predetermined lower limit value. For example, if the target average current I_{p2ave}^* is small, the controller **90** preferably reduces the number of pulses. By reducing the number of pulses, the pulse width is increased accordingly for applying the pump current I_{p2} according to the target average current I_{p2ave}^* , so that the pulse width can be prevented from decreasing. In the example in FIG. **9**, the pulse width of the waveform in the upper section is small such that the average value I_{p2ave} and the target average current I_{p2ave}^* deviate from each other. However, by using the waveform in the lower section as an alternative, the average value I_{p2ave} and the target average current I_{p2ave}^* are less likely to deviate from each other. FIG. **9** illustrates a case where the pump current I_{p2} is a burst pulse current, similar to FIG. **6**, and the waveform in the upper section has six pulses per cycle, whereas the number of pulses in the waveform in the lower section is reduced to two. Instead, the pulse width (i.e., the time T_a per pulse) of the waveform in the lower section of FIG. **9** is three times that of the waveform in the upper section, so that the ideal average value I_{p2ave} is not varied between the waveform in the upper section and the waveform in the lower section. In the waveform in the lower section of FIG. **9**, the burst-pulse interval (T_b in FIG. **9**) is increased (specifically by a factor of 3). The following description relates to an example of a process performed by the controller **90** in a case where the number of pulses of the pump current I_{p2} is reduced when the target average current I_{p2ave}^* is small. For example, the controller **90** repeatedly executes a process involving setting the target average current I_{p2ave}^* by performing feedback control based on the voltage V_2 and the target value V_2^* , and controlling the measurement pump cell **41** so as to set the average value I_{p2ave} to the set target average current I_{p2ave}^* . In this case, if the value of the set target average current I_{p2ave}^* is a value included in a predetermined low current range (e.g., a value smaller than a predetermined threshold value), the measurement pump cell **41** is controlled such that a pump current I_{p2} with a reduced number of pulses is applied, as compared with a case where the value of the set target average current I_{p2ave}^* is a value not included in the predetermined low current range (e.g., a value larger than or equal to the predetermined threshold value). FIG. **10** illustrates an example of comparison results between a case where control for reducing the number of pulses is not performed (i.e., the table at the left side) and a case where such control is performed (i.e., the table at the right side). FIG. **10** shows the height, width, and number of pulses of the pump current I_{p2} set by the controller **90** such that the average value I_{p2ave} becomes the target average current I_{p2ave}^* in a case where the target average current I_{p2ave}^* is varied between $0.06\ \mu\text{A}$ and $6\ \mu\text{A}$. The pump current I_{p2} is a burst pulse current, similar to FIGS. **6** and **9**, the pulse height corresponds to I_{p2max} in FIGS. **6** and **9**, the width corresponds to T_a in FIGS. **6** and **9**, and the number of pulses corresponds to the number of pulses per cycle.

Moreover, the presence and absence of a deviation between the actual average value I_{p2ave} and the target average current I_{p2ave}^* in a case where the pump current I_{p2} with the set pulse height, width, and number is applied is also shown. As shown at the left side of FIG. 10, in a case where the control for reducing the number of pulses is not performed and the number of pulses is fixed at 3, the set pulse width is a small value below 10 μs when the target average current I_{p2ave}^* is lower than 0.15 μA . The actual average value I_{p2ave} in this case is smaller than the ideal average value I_{p2ave} and deviates from the target average current I_{p2ave}^* . In contrast, in the example at the right side of FIG. 10, the controller 90 has performed the control for reducing the number of pulses to 1 when the target average current I_{p2ave}^* is lower than 0.15 μA . As a result, when the target average current I_{p2ave}^* is lower than 0.15 μA , the set number of pulses is reduced to 1, so that the pulse width is set to a value larger than or equal to 10 μs as a value required for achieving the target average current I_{p2ave}^* . Specifically, the pulse width is set to a larger value than that in the table at the left side of FIG. 10, and the pulse width does not fall below 10 μs . As a result, even when the target average current I_{p2ave}^* is lower than 0.15 μA , a deviation has not occurred between the actual average value I_{p2ave} and the target average current I_{p2ave}^* . In FIG. 10, the NO concentration corresponding to the target average current I_{p2ave}^* is also shown. In the example at the left side of FIG. 10, the actual average value I_{p2ave} deviates from the target average current I_{p2ave}^* when the target average current I_{p2ave}^* is lower than 0.15 μA , that is, when the NO concentration is a low concentration below 50 ppm, thus making it difficult to properly control the measurement pump cell 41, as mentioned above, or causing the S/N ratio of the pump current I_{p2} to decrease. In contrast, in the example at the right side of FIG. 10, such a problem is less likely to occur even when the NO concentration is a low concentration below 50 ppm.

(90) As mentioned above, the controller 90 preferably controls the pump current I_{p2} in a range where the pulse width of the pump current I_{p2} does not fall below the predetermined lower limit value. Likewise, the controller 90 preferably controls the pump current I_{p2} in a range where the pulse interval (T_b in FIG. 9) of the pump current I_{p2} does not fall below a predetermined lower limit value. This is because, in a case where the pulse interval is too small, a subsequent pulse may rise before the pulse current completely falls, thus causing a problem similar to the case where the pulse width is too small. In the examples in FIGS. 9 and 10, the pump current I_{p2} is described as being a burst pulse current. Even in a case where the pump current I_{p2} is not a burst pulse current, as in FIG. 3, it is preferable that the pump current I_{p2} be controlled in a range where the pulse width (T_{on} in FIG. 3) and the pulse interval (T_{off} in FIG. 3) do not fall below the predetermined lower limit values.

(91) Although the controller 90 and the power sources 24, 46, and 52 are illustrated separately from each other in FIG. 2 in the above-described embodiment, the power sources 24, 46, and 52 may be regarded as being a part of the controller 90.

EXAMPLES

(92) Specific fabrication examples of gas sensors will be described below as practical examples. The present invention is not to be limited to the following practical examples.

Practical Example 1

(93) Practical Example 1 is achieved by fabricating the gas sensor 100 shown in FIGS. 1 and 2. The sensor element 101 is fabricated as follows. First, ceramic green sheets corresponding to the individual layers 1 to 6 are each formed by mixing zirconia particles having 4 mol % of yttria added thereto as a stabilizer with an organic binder, a dispersant, a plasticizer, and an organic solvent, and then molding the mixture by tape molding. Then, the ceramic green sheets corresponding to the individual layers undergo hole-drilling, where appropriate, and screen-printing of a pattern of a conductive paste for forming electrodes and a circuit, and are subsequently stacked and pressure-bonded, so that a pressure-bonded body is obtained. A pattern for the measurement electrode 44 is formed by screen-printing a conductive paste having a mixture of zirconia particles and an organic binder onto Pt and Rh serving as noble metals. Then, an unbaked layered body

having the size of the sensor element **101** is cut out from the pressure-bonded body and is baked, whereby the sensor element **101** is obtained. The fabricated sensor element **101** is electrically connected with the power sources and the controller **90**, whereby the gas sensor **100** according to Practical Example 1 is fabricated. The pump current I_{p2} to be applied by the pulse power source **46** is a rectangular single pulse current shown in FIG. **3**, has a frequency of 100 Hz (i.e. a cycle T of 0.1 s), and a maximum current I_{p2max} of 50 μ A.

Practical Example 2

(94) In Practical Example 2, a gas sensor **100** similar to that in Practical Example 1 is fabricated except that the pump current I_{p2} to be applied by the pulse power source **46** is changed. The pump current I_{p2} to be applied by the pulse power source **46** is a burst pulse current shown in FIG. **6**, has a frequency of 100 Hz (i.e. a cycle T of 0.1 s), has five pulses per cycle, and has 50% as the proportion (i.e., duty ratio) of the oscillation period T_A occupying the cycle T (burst cycle).

Practical Example 3

(95) In Practical Example 3, a gas sensor **100** similar to that in Practical Example 1 is fabricated except that the frequency is set to 200 Hz (i.e., the cycle T is set to 0.05 s).

Practical Example 4

(96) In Practical Example 4, a gas sensor **100** similar to that in Practical Example 2 is fabricated except that the frequency is set to 200 Hz (i.e., the cycle T is set to 0.05 s).

Comparative Example 1

(97) In Comparative Example 1, the gas sensor **900** according to the comparative example shown in FIG. **4** is fabricated. Specifically, in Comparative Example 1, the pump current I_{p2} to be applied is a continuous electric current. The gas sensor **900** according to Comparative Example 1 is similar to that in Practical Example 1 except that the gas sensor **900** includes the measurement pump cell **941** shown in FIG. **4** in place of the measurement pump cell **41**.

(98) [Durability Test]

(99) A durability test using a diesel engine is performed with respect to each of Practical Examples 1 to 4 and Comparative Example 1, and the degree of degradation in the measurement electrode **44** is evaluated. As indicators for evaluating the degree of degradation, the NO-sensitivity change rate before and after the durability test and the NO-sensitivity linear change rate before and after the durability test are measured. In detail, the test is performed as follows. The gas sensor according to Practical Example 1 is attached to an exhaust gas pipe of an automobile. Then, the temperature is set to 800° C. by applying electricity to the heater **72**, thereby heating the sensor element **101**. The controller **90** controls the aforementioned pump cells **21**, **41**, and **50** and acquires the voltages V_0 , V_1 , V_2 , and V_{ref} from the aforementioned sensor cells **80** to **83**. In this state, a first model gas having nitrogen as its base gas and having a NO concentration of 500 ppm is made to flow through the pipe. After waiting for a control value from the controller **90** to the pulse power source **46** to become stable, the average value I_{p2ave} is measured as an initial value I_{a500} of the detection sensitivity of the gas sensor relative to the NO at 500 ppm. Subsequently, a 40-minute operation pattern with an engine rotation speed ranging from 1500 to 3500 rpm and a load torque ranging from 0 to 350 N.Math.m is repeated as a durability test for 1000 hours. In this case, the gas temperature is between 200° C. and 600° C., and the NOx concentration is between 0 ppm and 1500 ppm. During this 1000-hour period, the aforementioned individual pump cells are being continuously controlled and the voltages are being continuously acquired by the controller **90**. After the lapse of 1000 hours, the average value I_{p2ave} is measured by using a method similar to that for the initial value I_{a500} , and is set as a post-durability-test value I_{b500} . Then, the NO-sensitivity change rate [%] of the pump current I_{p2} before and after the durability test performed on the gas sensor according to Practical Example 1 is determined as the sensitivity change rate at the NO concentration of 500 ppm = $[1 - (\text{post-durability-test value } I_{b500} / \text{initial value } I_{a500})] \times 100\%$. Moreover, the NO-sensitivity linear rate after the durability test performed on the gas sensor according to Practical Example 1 is determined as follows. First, similar to the post-durability-test

value Ib500, the average value Ip2ave after a control value has become stable is measured with respect to a second model gas having nitrogen as its base gas and having an NO concentration of 0 ppm, and is set as a post-durability-test value Ib0. Moreover, the average value Ip2ave after a control value has become stable is also measured with respect to a third model gas having nitrogen as its base gas and having an NO concentration of 1500 ppm, and is set as a post-durability-test value Ib1500. Then, an NO-sensitivity gradient A between two points respectively corresponding to the case where the NO concentration in the gas sensor according to Practical Example 1 after the durability test is 0 ppm and the case where the NO concentration is 500 ppm is determined as a gradient $A = (Ib500 - Ib0) / (500 - 0)$. Moreover, an NO-sensitivity gradient B between two points respectively corresponding to the case where the NO concentration in the gas sensor according to Practical Example 1 after the durability test is 0 ppm and the case where the NO concentration is 1500 ppm is determined as a gradient $B = (Ib1500 - Ib0) / (1500 - 0)$. Then, the NO-sensitivity linear rate [%] after the durability test is determined as a NO-sensitivity linear rate = (gradient B) / (gradient A) × 100%. For Practical Examples 2 to 4, the NO-sensitivity change rate [%] of the pump current Ip2 before and after the durability test and the NO-sensitivity linear rate [%] after the durability test are similarly determined. For Comparative Example 1, the NO-sensitivity change rate [%] and the NO-sensitivity linear rate [%] are similarly determined except that the value of the pump current Ip2 after being stable is measured in place of the average value Ip2ave.

(100) Table 1 shows the frequency, the pulse method, the NO-sensitivity change rate, and the NO-sensitivity linear rate with respect to Practical Examples 1 to 4 and Comparative Example 1. A change in the pump current Ip2 relative to the NO between the initial value and the state after the durability test decreases with decreasing absolute value of the NO-sensitivity change rate. This implies that deterioration in the detection accuracy with use of the gas sensor is suppressed.

Furthermore, as the NO-sensitivity linear rate after the durability test approaches 100%, the NO-sensitivity linearity between three points respectively corresponding to 0 ppm, 500 ppm, and 1500 ppm as the NO concentration values is closer to an ideal state even after the durability test. This implies that deterioration in the detection accuracy with use of the gas sensor is suppressed. It is clear from Table 1 that, with regard to both the NO-sensitivity change rate and the NO-sensitivity linear rate, the results of Practical Examples 1 to 4 are more favorable than that of Comparative Example 1, and deterioration in the detection accuracy with use of the gas sensor is suppressed. This is conceivably because a decrease in the catalytic activity of the measurement electrode 44 caused by the pump current Ip2 flowing thereto is suppressed in Practical Examples 1 to 4 due to the aforementioned reasons. In each of Practical Examples 1 to 4 and Comparative Example 1, the NO-sensitivity linear rate measured before the durability test is approximately 98%.

(101) TABLE-US-00001

TABLE 1	NO-sensitivity change rate	linear before and change rate after the after the durability durability	Frequency	Pulse method	test test
Experimental Example 1	100	Rectangular single pulse	-5.1%	94.0%	
Experimental Example 2	100	Burst pulse	-5.8%	94.2%	
Experimental Example 3	200	Rectangular single pulse	-5.7%	93.5%	
Experimental Example 4	200	Burst pulse	-6.0%	93.8%	
Comparative Example 1	—	—	-10.2%	89.6%	

Claims

1. A gas sensor that detects a specific gas concentration in a measurement-object gas, the gas sensor comprising: an element body including an oxygen-ion-conductive solid electrolyte layer and provided with a measurement-object gas flow section therein, the measurement-object gas flow section receiving the measurement-object gas and allowing the measurement-object gas to flow therethrough; a main pump cell having an inner main pump electrode and an outer main pump electrode, the inner main pump electrode being disposed in a first internal cavity in the measurement-object gas flow section and containing a catalytically-active noble metal, the outer main pump electrode being disposed in an area to be exposed to the measurement-object gas at an

outer side of the element body, the main pump cell pumping out oxygen from the first internal cavity, an auxiliary pump cell having an inner auxiliary pump electrode and an outer auxiliary pump electrode, the inner auxiliary pump electrode being disposed in a second internal cavity provided downstream of the first internal cavity in the measurement-object gas flow section and containing a catalytically-active noble metal, the outer auxiliary pump electrode being disposed in an area to be exposed to the measurement-object gas at an outer side of the element body, the auxiliary pump cell pumping out oxygen from the second internal cavity, a measurement pump cell having an inner measurement electrode and an outer measurement electrode, the inner measurement electrode being disposed in a measurement chamber provided downstream of the second internal cavity in the measurement-object gas flow section and containing a catalytically-active noble metal, the outer measurement electrode being disposed in an area to be exposed to the measurement-object gas at an outer side of the element body, the measurement pump cell pumping out oxygen produced in the measurement chamber from the specific gas, a CPU coupled to a memory storing instructions that when executed by the CPU configure the CPU to: pump out oxygen from the measurement chamber by applying a measurement pump current repeatedly as an on-off controlled pump current between the inner measurement electrode and the outer measurement electrode; and a reference electrode that is disposed inside the element body and that receives a reference gas serving as a reference for detection of the specific gas concentration; wherein the CPU is configured to: detect a measurement voltage between the reference electrode and the inner measurement electrode, control the measurement pump current based on the measurement voltage during a second period among a first period and the second period so as to set an oxygen concentration in the measurement chamber to a predetermined low concentration, the first period being a period in which a change has occurred in the measurement voltage due to the measurement pump current being in an on mode, and the second period being a period in which the change in the measurement voltage has receded as compared with the first period due to the measurement pump current being in an off mode; and detect the specific gas concentration in the measurement-object gas based on the measurement pump current.

2. A gas sensor that detects a specific gas concentration in a measurement-object gas, the gas sensor comprising: an element body including an oxygen-ion-conductive solid electrolyte layer and provided with a measurement-object gas flow section therein, the measurement-object gas flow section receiving the measurement-object gas and allowing the measurement-object gas to flow therethrough; a main pump cell having an inner main pump electrode and an outer main pump electrode, the inner main pump electrode being disposed in a first internal cavity in the measurement-object gas flow section and containing a catalytically-active noble metal, the outer main pump electrode being disposed in an area to be exposed to the measurement-object gas at an outer side of the element body, the main pump cell pumping out oxygen from the first internal cavity, an auxiliary pump cell having an inner auxiliary pump electrode and an outer auxiliary pump electrode, the inner auxiliary pump electrode being disposed in a second internal cavity provided downstream of the first internal cavity in the measurement-object gas flow section and containing a catalytically-active noble metal, the outer auxiliary pump electrode being disposed in an area to be exposed to the measurement-object gas at an outer side of the element body, the auxiliary pump cell pumping out oxygen from the second internal cavity, a measurement pump cell having an inner measurement electrode and an outer measurement electrode, the inner measurement electrode being disposed in a measurement chamber provided downstream of the second internal cavity in the measurement-object gas flow section and containing a catalytically-active noble metal, the outer measurement electrode being disposed in an area to be exposed to the measurement-object gas at an outer side of the element body, the measurement pump cell pumping out oxygen produced in the measurement chamber from the specific gas, a CPU coupled to a memory storing instructions that when executed by the CPU configure the CPU to: pump out oxygen from the measurement chamber by applying a measurement pump current repeatedly as an

on-off controlled pump current between the inner measurement electrode and the outer measurement electrode wherein a first period and a second period occur as a result of the on-off controlled pump current, the first period being a period in which a change has occurred in a measurement voltage due to the measurement pump current being in an on mode, and the second period being a period in which the change in the measurement voltage has receded as compared with the first period due to the measurement pump current being in an off mode; and a reference electrode that is disposed inside the element body and that receives a reference gas serving as a reference for detection of the specific gas concentration; wherein the CPU is configured to: detect the measurement voltage between the reference electrode and the inner measurement electrode; control the measurement pump current so as to set an average value of the measurement pump current to a predetermined target value; and detect the specific gas concentration in the measurement-object gas based on the measurement voltage during only the second period.
